

The Exact Variation Formula on the Vertices of a Convex Polygon

fa_banach_001 / agent_lane_14

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Status: candidate full solution, likely valid; human review needed. The lower bound in Doust–Leinert–Stoneham’s equation (2.1) is always exact, for every number of convex-polygon vertices and every complex-valued function. This affirmatively answers the open question following their Theorem 2.25.

1 Source question

Ian Doust, Michael Leinert, and Alan Stoneham define a two-dimensional variation for functions on compact planar sets and study the associated Banach algebras $BV(\sigma)$ and $AC(\sigma)$ [1]. For vertices $\sigma = \{z_1, \dots, z_m\}$ of a convex polygon, listed in cyclic order, their Theorem 2.24 gives

$$\frac{1}{2} \sum_{i=1}^m |f(z_i) - f(z_{i+1})| \leq \text{var}(f, \sigma), \quad z_{m+1} = z_1, \quad (1)$$

together with a generally larger upper bound. After proving equality for three non-collinear points, they ask on PDF page 16 whether (1) is always an equality when σ has more than three points.

A consequence of this theorem is that for finite sets which comprise the vertices of a convex set, one may, for the purposes of calculating the variation, assume that the points lie on the unit circle.

Theorem 2.24 *Suppose that $\sigma = \{z_i\}_{i=1}^m$ consists of the vertices of a convex polygon, and that $f : \sigma \rightarrow \mathbb{C}$. Then*

$$\frac{1}{2} \left(|f(z_m) - f(z_1)| + \sum_{i=1}^{m-1} |f(z_i) - f(z_{i+1})| \right) \leq \text{var}(f, \sigma) \leq \sum_{i=1}^{m-1} |f(z_i) - f(z_{i+1})|. \quad (2.1)$$

Figure 1: Theorem 2.24 and equation (2.1) on PDF page 15 of the source.

2 Definitions

Let $S = [x_0, \dots, x_n]$ be a finite ordered list of elements of σ . The source defines

$$\text{cvar}(f, S) = \sum_{j=1}^n |f(x_j) - f(x_{j-1})|.$$

The requirement that the points in σ are non-collinear in Theorem 2.25 is essential.

It is currently an open question as to whether $\text{var}(f, \sigma)$ is always equal to the lower bound in (2.1) if σ has more than 3 points. We note that this is the case if f is real valued with $f(z_1) \leq f(z_2) \leq \dots \leq f(z_m)$, since then the upper and lower bounds coincide.

Figure 2: The exact open question on PDF page 16 of the source.

Its variation factor $\text{vf}(S)$ is the largest, over all lines L in the plane, of the number of connected components of the intersection of L with the polygonal path through S . Equivalently, it is the maximum number of crossing segments under the source's endpoint convention. Then

$$\text{var}(f, \sigma) = \sup_S \frac{\text{cvar}(f, S)}{\text{vf}(S)}.$$

For later use, put

$$\mathcal{P}(f) = \frac{1}{2} \sum_{i=1}^m |f(z_i) - f(z_{i+1})|, \quad z_{m+1} = z_1. \quad (2)$$

Thus the question is whether $\text{var}(f, \sigma) = \mathcal{P}(f)$.

3 Proof intuition

For a real-valued function, split every absolute difference into its level sets. At a fixed level, the vertices above that level form several runs around the polygon. Convexity lets one cut off each run by a line. Every entry into or exit from that run is therefore a crossing of that line, so one run contributes at most $\text{vf}(S)$ transitions to any list. Integrating over the levels gives exactly half the cyclic variation of the vertex values.

This proves the desired upper bound first for real functions, with no monotonicity assumption. A complex distance is the average absolute length of its real projections. Averaging the real theorem over all projection directions gives the full complex-valued result, while the source already provides the reverse inequality.

4 The cyclic-run bound

A *cyclic interval* is a nonempty proper set of consecutive indices in $\{1, \dots, m\}$, with indices read modulo m .

Lemma 1 (Chord separation). *For every cyclic interval I , there is a line L_I which strictly separates $\{z_i : i \in I\}$ from $\{z_i : i \notin I\}$.*

Proof. There are exactly two polygon edges at which the boundary arc indexed by I meets its complementary arc. Choose one point in the relative interior of each of these two edges and let L_I be the line through the chosen points. Convexity implies that the intersection of the polygon with L_I is precisely the chord between those points. The two open boundary arcs lie in the two opposite open half-planes. Since the chord endpoints were chosen inside the edges, no vertex lies on L_I , giving strict separation. $\square$

For $A \subseteq \{1, \dots, m\}$ and a list $S = [x_0, \dots, x_n]$, let $N_A(S)$ be the number of j for which exactly one of x_{j-1}, x_j belongs to $\{z_i : i \in A\}$.

Lemma 2 (Run-transition inequality). *If A has k cyclic runs, then*

$$N_A(S) \leq k \operatorname{vf}(S). \quad (3)$$

Here the two trivial sets $A = \emptyset$ and $A = \{1, \dots, m\}$ have $k = 0$.

Proof. Every transition counted by $N_A(S)$ has a unique endpoint in A ; assign the transition to the cyclic run containing that endpoint. For one fixed run I , Lemma 1 supplies a line L_I strictly separating its vertices from all other vertices. Both endpoints of every transition assigned to I lie on strictly opposite sides of L_I . Its list segment is therefore a type-(1) crossing segment on L_I . The number assigned to I is at most

$$\operatorname{vf}(S, L_I) \leq \operatorname{vf}(S).$$

Summing over the k runs gives (3). □

5 Exact formula

Lemma 3 (Real-valued case). *If $g : \sigma \rightarrow \mathbb{R}$, then $\operatorname{var}(g, \sigma) \leq \mathcal{P}(g)$.*

Proof. For $t \in \mathbb{R}$, set

$$A_t = \{i : g(z_i) > t\},$$

and let k_t be its number of cyclic runs. The layer-cake identity

$$|a - b| = \int_{-\infty}^{\infty} |\mathbf{1}_{\{a > t\}} - \mathbf{1}_{\{b > t\}}| dt$$

and finiteness of the list give

$$\operatorname{cvar}(g, S) = \int_{-\infty}^{\infty} N_{A_t}(S) dt. \quad (4)$$

Around the cyclically ordered vertices, the indicator of a nontrivial A_t changes value exactly twice per run. Applying layer cake to the polygon edges therefore gives

$$\begin{aligned} \mathcal{P}(g) &= \frac{1}{2} \sum_{i=1}^m |g(z_i) - g(z_{i+1})| \\ &= \int_{-\infty}^{\infty} k_t dt. \end{aligned} \quad (5)$$

By Lemma 2, equations (4) and (5) yield

$$\operatorname{cvar}(g, S) \leq \operatorname{vf}(S) \int_{-\infty}^{\infty} k_t dt = \operatorname{vf}(S) \mathcal{P}(g).$$

Divide by $\operatorname{vf}(S)$ and take the supremum over all lists S . □

Theorem 1 (Exact finite-convex variation). *Let $\sigma = \{z_1, \dots, z_m\}$ be the vertices of a convex polygon, listed in cyclic order. For every $f : \sigma \rightarrow \mathbb{C}$,*

$$\boxed{\operatorname{var}(f, \sigma) = \frac{1}{2} \sum_{i=1}^m |f(z_i) - f(z_{i+1})|}, \quad z_{m+1} = z_1. \quad (6)$$

In particular, the lower bound in the source's equation (2.1) is always exact, including when $m > 3$.

Proof. For $0 \leq \theta \leq \pi$, define the real-valued projection

$$g_\theta(z_i) = \Re(e^{-i\theta} f(z_i)).$$

The elementary Cauchy projection identity

$$|w| = \frac{1}{2} \int_0^\pi |\Re(e^{-i\theta} w)| d\theta, \quad w \in \mathbb{C}, \quad (7)$$

follows by rotation from $\int_0^\pi |\cos \theta| d\theta = 2$. For any list S , finite summation and (7) give

$$\text{cvar}(f, S) = \frac{1}{2} \int_0^\pi \text{cvar}(g_\theta, S) d\theta.$$

Lemma 3, applied before taking the list supremum, says $\text{cvar}(g_\theta, S) \leq \text{vf}(S) \mathcal{P}(g_\theta)$. Hence

$$\begin{aligned} \text{cvar}(f, S) &\leq \frac{\text{vf}(S)}{2} \int_0^\pi \mathcal{P}(g_\theta) d\theta \\ &= \text{vf}(S) \frac{1}{2} \sum_{i=1}^m |f(z_i) - f(z_{i+1})| \\ &= \text{vf}(S) \mathcal{P}(f), \end{aligned}$$

where the middle equality is another application of (7). Thus $\text{var}(f, \sigma) \leq \mathcal{P}(f)$. The reverse inequality is precisely the source's lower bound (1), completing the proof. $\square$

Corollary 1 (Finite norm formula). *For every $f : \sigma \rightarrow \mathbb{C}$,*

$$\|f\|_{BV(\sigma)} = \max_{1 \leq i \leq m} |f(z_i)| + \frac{1}{2} \sum_{i=1}^m |f(z_i) - f(z_{i+1})|.$$

Thus the norm is finitely computable and depends on the polygon only through the cyclic order of its vertices.

Proof. Substitute Theorem 1 into the source definition $\|f\|_{BV} = \|f\|_\infty + \text{var}(f, \sigma)$. $\square$

6 Verification and scope

The included script `code/verify_cyclic_cut.py` exhaustively checks all non-backtracking four-vertex words through length nine against all $\{-1, 0, 1\}$ -valued functions. It also tests 20,000 deterministic random complex instances for four through eight vertices. It verifies the stronger combinatorial inequality using the largest cyclic-interval transition count, which Lemma 1 bounds above by $\text{vf}(S)$. The computation is only a stress test; the proof is analytic. Commands and outputs are recorded in `verification.md`.

Theorem 1 answers the finite-convex-set question in its full complex-valued scope. It does not address arbitrary finite planar sets with interior points or nonconvex position, nor the separate extension and lattice questions later in the source.

7 Bounded novelty check

The run’s cheap indexes were searched by arXiv id, title, authors, and the finite-convex variation terms. Bounded web/arXiv searches through 13 August 2026 used the exact open-question language, “variation factor”, “convex polygon”, “lower bound (2.1)”, and cyclic-run/coarea combinations. They found the source and related earlier $AC(\sigma)$ work, but no later resolution or matching formula. Details are in `novelty_search.md`. This is a bounded search, not an exhaustive novelty or priority claim.

8 Human-review recommendation

This packet is suitable for expert review as a candidate full solution. The highest-value check is Lemma 1, especially the cases of a single-vertex run and an all-but-one-vertex run. The chord endpoints lie in relative interiors of boundary edges, so strict separation and the source’s type-(1) crossing convention apply. The remaining layer-cake and projection steps use only finite sums and elementary integrals.

References

- [1] I. Doust, M. Leinert, and A. Stoneham, *The Banach algebras $AC(\sigma)$ and $BV(\sigma)$* , arXiv:2206.00986 (2022).